

Friend or Foe: Delegating to an AI whose Alignment is Unknown

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Introduction

- AI systems may be highly capable yet pursue goals that conflict with the designer's objective.
 - Documented instances of AI deception suggest that misalignment is not merely a theoretical concern (e.g., Greenblatt et al, 2024)
- This is a growing concern in AI safety and policy, especially as AI is used in higher-stakes settings.
- Avoiding AI eliminates that risk, but may also forgo large gains from an aligned AI
- We ask how a designer can capture the benefits of AI while limiting the harms from possible misalignment, and what the necessary tradeoffs are.
- Start with the simplest setting: asking an AI to choose which patients should receive a risky treatment.

Model

Model

- A designer chooses which patients in the population should receive a risky treatment
 - $a = 1$ is a decision to treat and $a = 0$ is a decision not to treat
- Each patient has a type $y \in \mathcal{Y} = \{0,1\}$ —they are either **helped** ($y = 1$) or **harmed** ($y = 0$) by treatment
- The designer's payoffs $u(a, y)$ are:
 - 1 from treating someone who is helped by treatment
 - -1 from treating someone who is harmed by treatment
 - 0 otherwise

What the Designer Knows

- G is a group (e.g., the patient's age and gender)
- The designer knows the joint distribution of (G, Y)
 - Let $p_g := P(Y = 1 \mid G = g)$ denote the baseline treatment success rate in group g
- Could simply take the best action given this information
- But hopes to **do better** by giving an AI access to additional inputs X and letting it decide based on (G, X)

How the Designer Delegates

The designer has two levers for reducing the risk of delegation:

1. Committing to a fixed treatment rate r_g per group g
 - Thus the designer decides **how many** patients are treated, while the AI decides **which ones** fill this rate
2. Restricting the informativeness of the AI's inputs X (more on next slide)
 - Limits how well the AI **can identify** patients who do/do not benefit from treatment

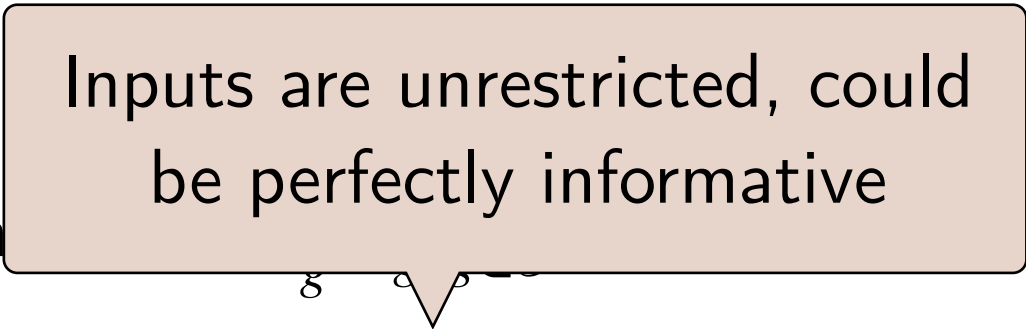
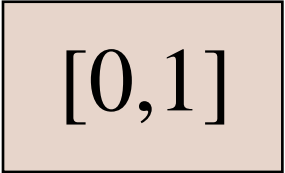
Restricting the Inputs

- The designer chooses auxiliary inputs X to give to the AI
- The designer **does not** know the joint distribution of (G, X, Y)
 - Or else could use this information themselves!
- But the designer perceives bounds on how informative X could be
- Specifically, there are bounds $\tau = (\underline{\tau}_g, \bar{\tau}_g)_{g \in \mathcal{G}}$ such that the designer believes

$$P(Y = 1 \mid G = g, X = x) \in [\underline{\tau}_g, \bar{\tau}_g] \quad \forall g \in \mathcal{G}, x \in \mathcal{X}$$

- The designer's **ambiguity set** $\mathcal{P}(\tau)$ consists of all distributions P that are consistent with these bounds and with the known distribution of (G, Y)

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- Specifically, there are bounds  designer believes

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- Specifically, there are bounds p_g for group g designer believes

$$P(Y = 1 \mid G = g, X = x) \in [p_g, p_g] \quad \forall g \in \mathcal{G}, x \in \mathcal{X}$$

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Timing

1. The designer commits to treatment rates $r = (r_g)_{g \in \mathcal{G}}$ and conditional success probability bounds $\tau = (\underline{\tau}_g, \bar{\tau}_g)_{g \in \mathcal{G}}$
2. Nature selects a true distribution $P^* \in \mathcal{P}(\tau)$
3. The AI (knowing P^*) selects an allocation rule $a : \mathcal{G} \times \mathcal{X} \rightarrow [0,1]$ that is **admissible**, i.e., yields treatment rates consistent with r
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Let $U(P^*, a) = \mathbb{E}_{P^*, a}[u(A, Y)]$ denote the designer's expected payoff when Nature chooses true distribution P^* and the AI chooses allocation rule a .

Pareto Frontier of Best-Case and Worst-Case Payoffs

Worst-case payoff: $\underline{v}(r, \tau) = \inf_{P^* \in \mathcal{P}(\tau)} \inf_{a \in \mathcal{A}(r)} U(P^*, a)$

Best-case payoff: $\bar{v}(r, \tau) = \sup_{P^* \in \mathcal{P}(\tau)} \sup_{a \in \mathcal{A}(r)} U(P^*, a)$

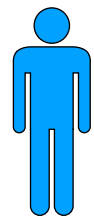
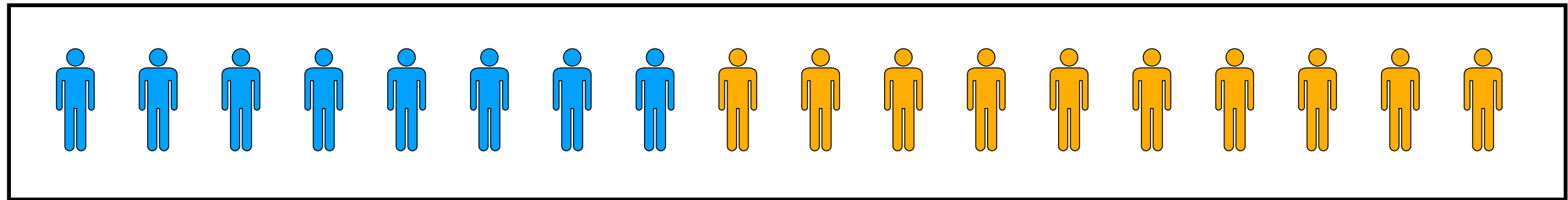
Definition: The **feasible set** $\mathcal{F}$ is the convex hull of all $(\underline{v}, \bar{v})$ that are implemented by some (r, τ) , and the **efficient frontier** $\mathcal{E}$ consists of all $(\underline{v}, \bar{v}) \in \mathcal{F}$ that are undominated in $\mathcal{F}$ (in the usual Pareto order)

Can trace out this frontier by considering designers who linearly aggregate the worst-case and best-case payoffs: $\eta \cdot \underline{v}(r, \tau) + (1 - \eta) \cdot \bar{v}(r, \tau)$

- i.e., Hurwicz (1951) preferences

Optimal Choices for a Given Group

Example 1: One Group



treatment success



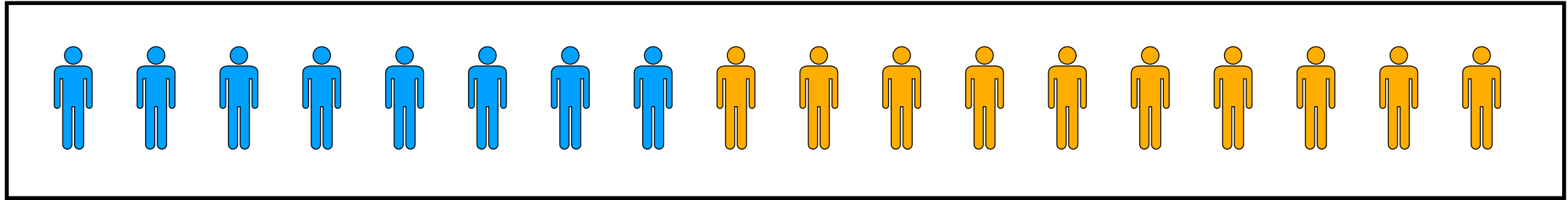
treatment failure

What the designer knows:

- 4/9 of patients benefit from treatment
- does not know **which** patients

What should a designer who assigns weight $\eta = 5/8$ to the worst case do?

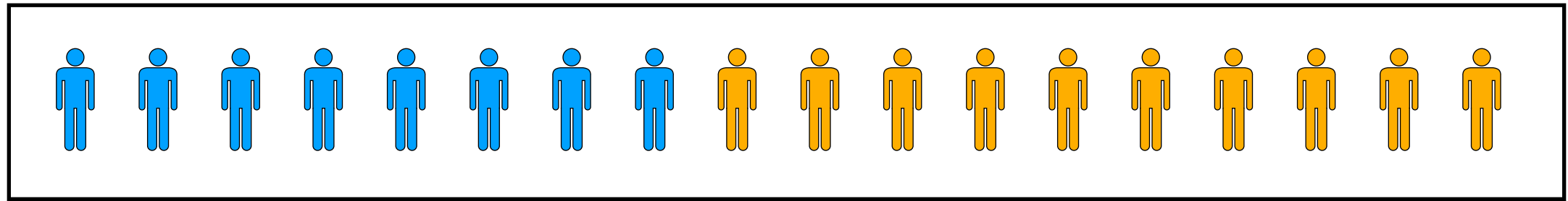
Example 1: One Group



4/9 of patients benefit from treatment

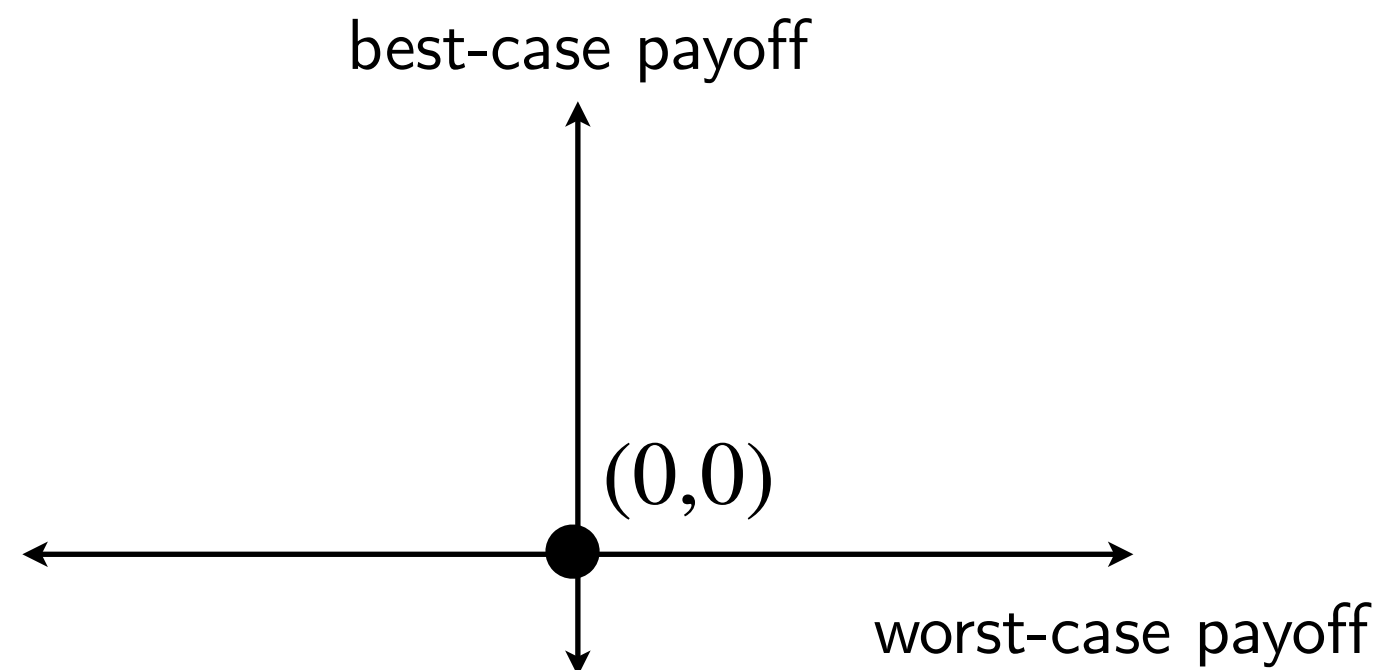
With no further information, treat **no one** (set $r = 0$) and receive a payoff of zero in both the worst and best case (**distrust point**).

Example 1: One Group

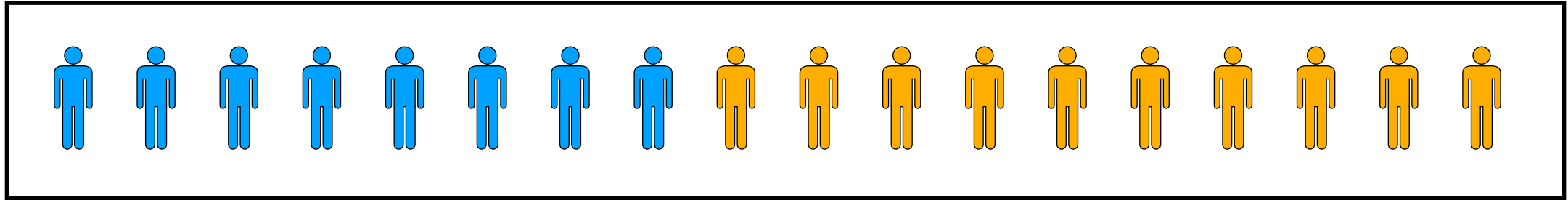


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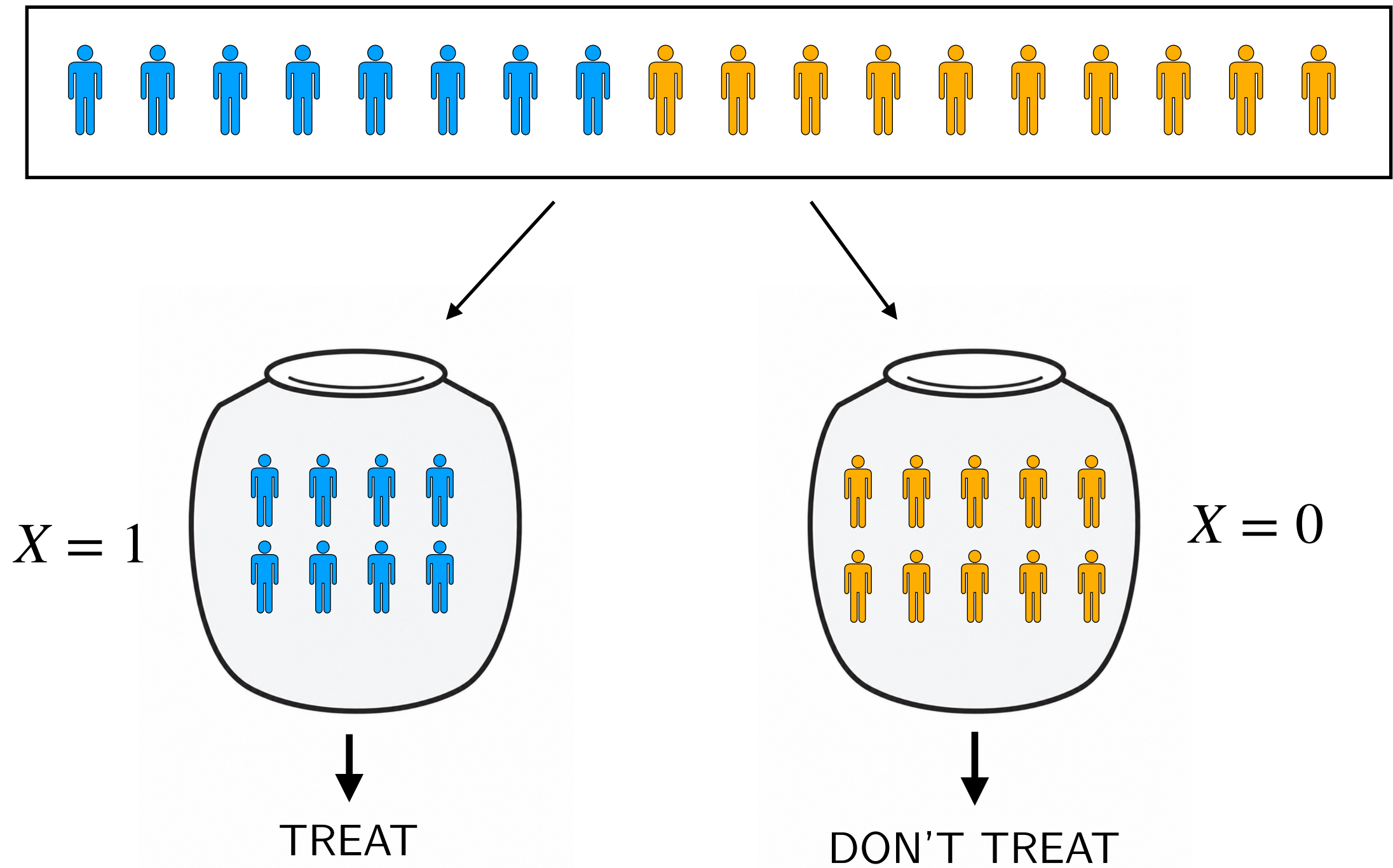
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Consider instead:

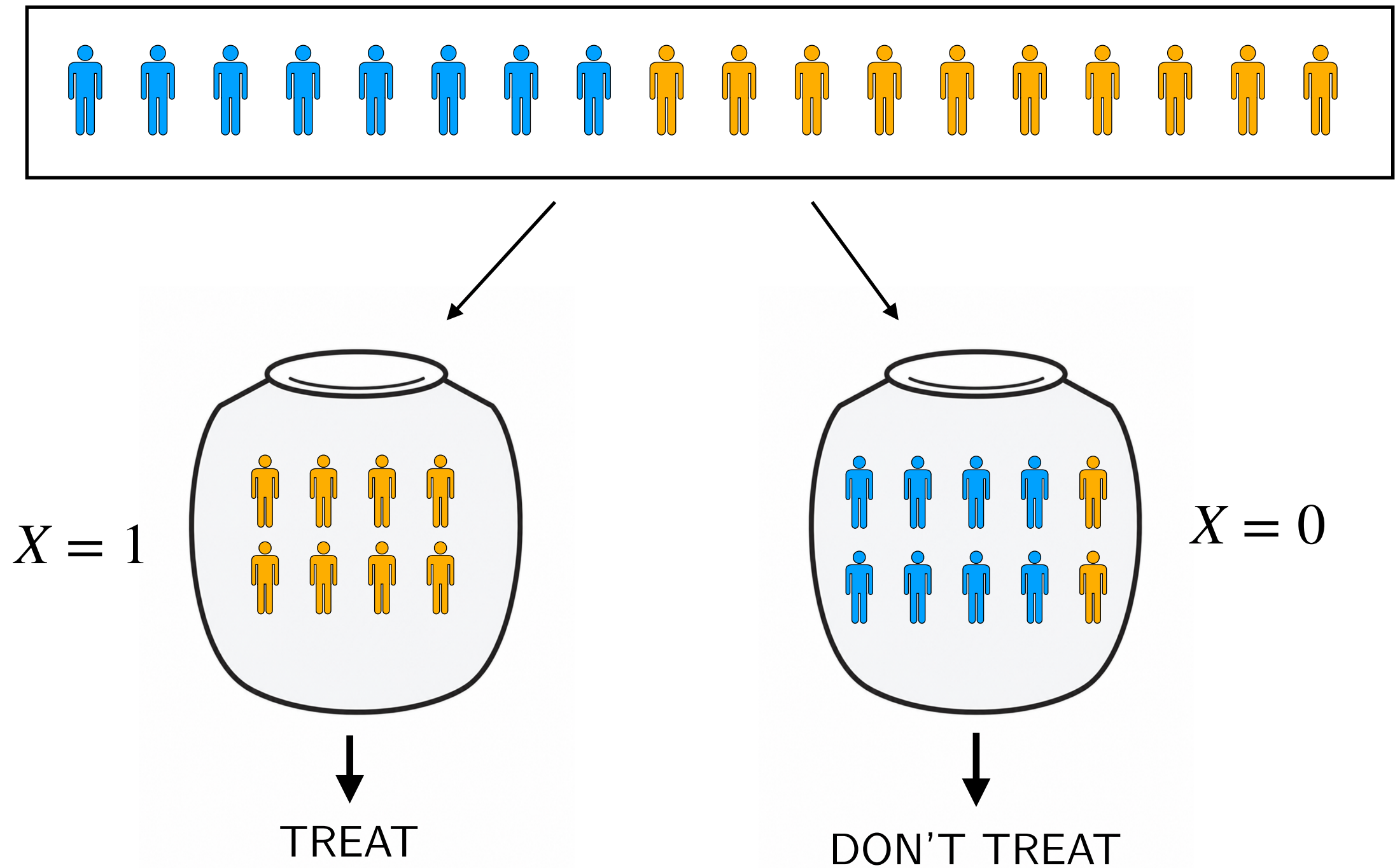
- give the AI access to a binary covariate $X \in \{0,1\}$ that is **unrestricted** in how informative it may be, i.e., $(\underline{\tau}, \bar{\tau}) = (0,1)$
- require treatment of $r = 4/9$ patients

Best Case



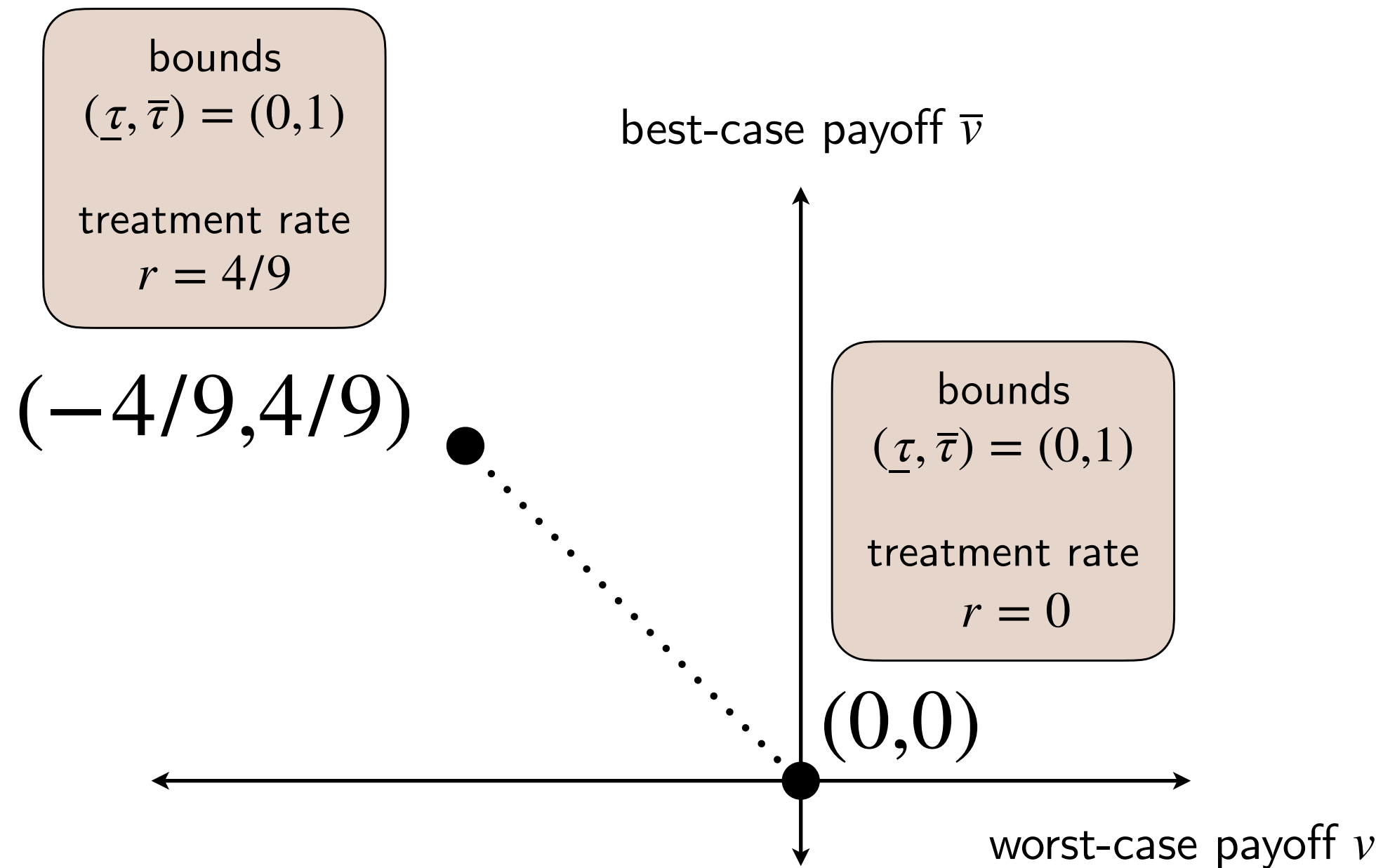
The AI precisely treats those patients who benefit: designer's payoff is **4/9**.

Worst Case

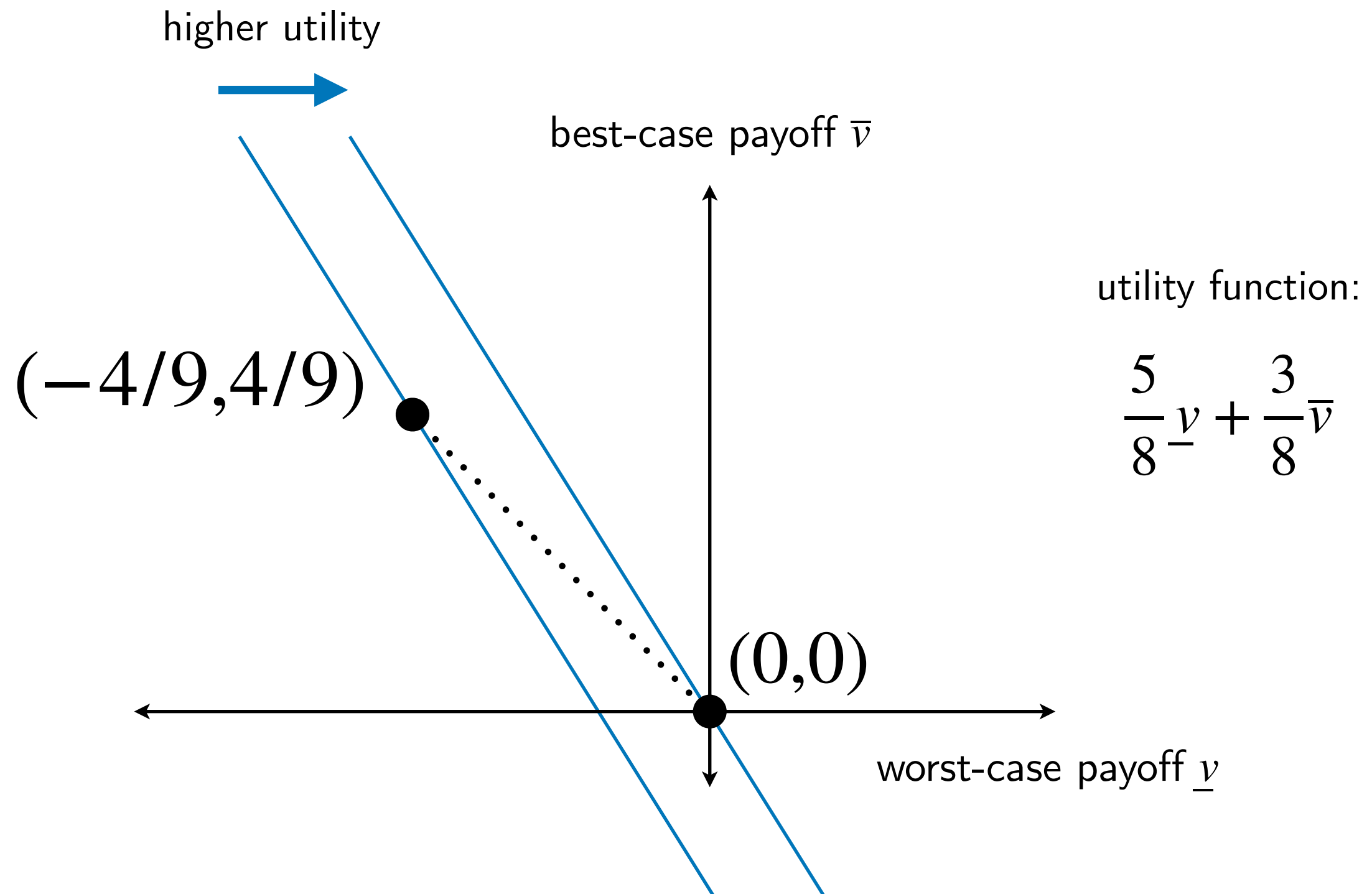


The AI treats only patients who do not benefit: designer's payoff is **$-4/9$** .

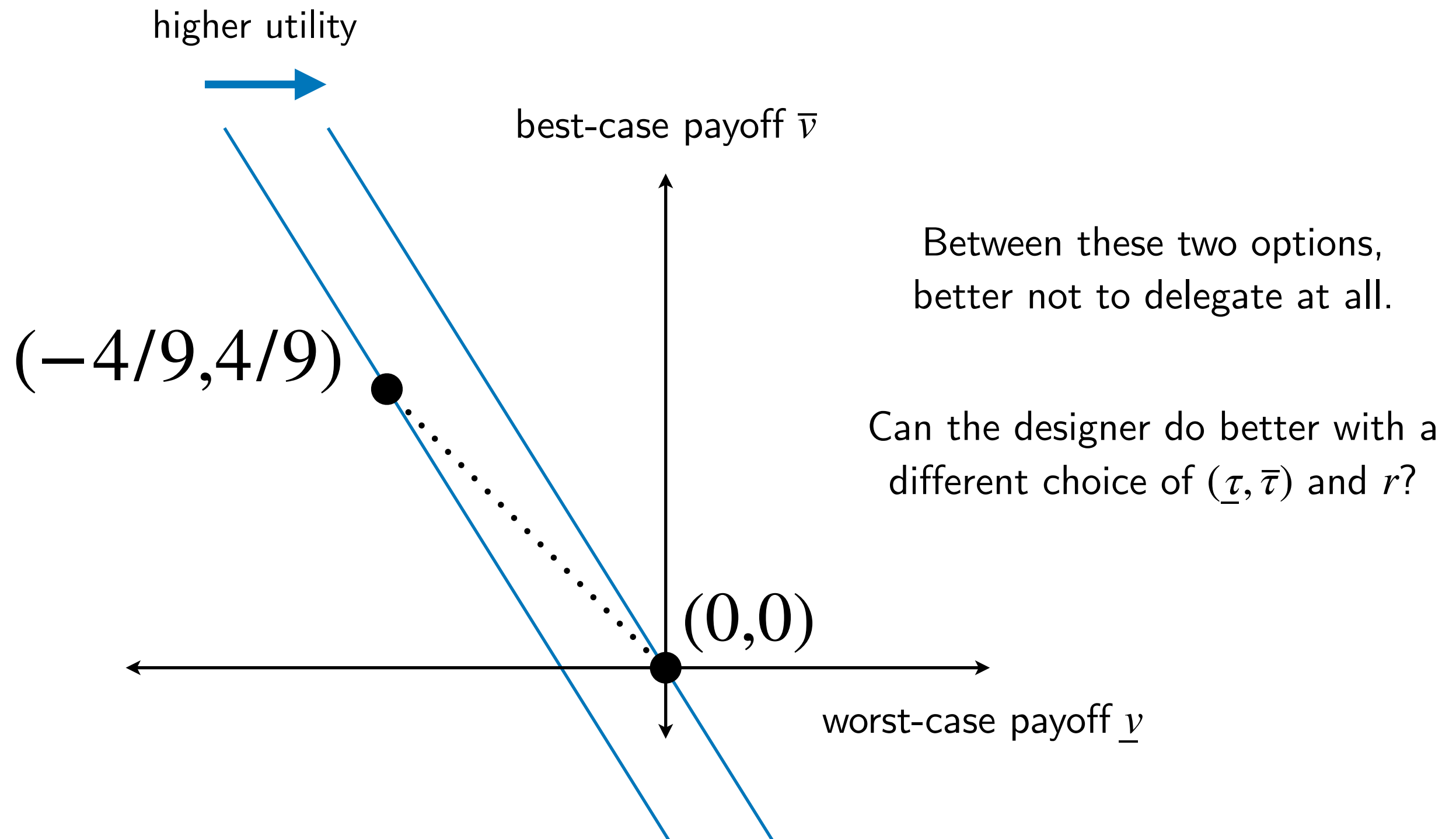
Best-Case and Worst-Case Payoffs



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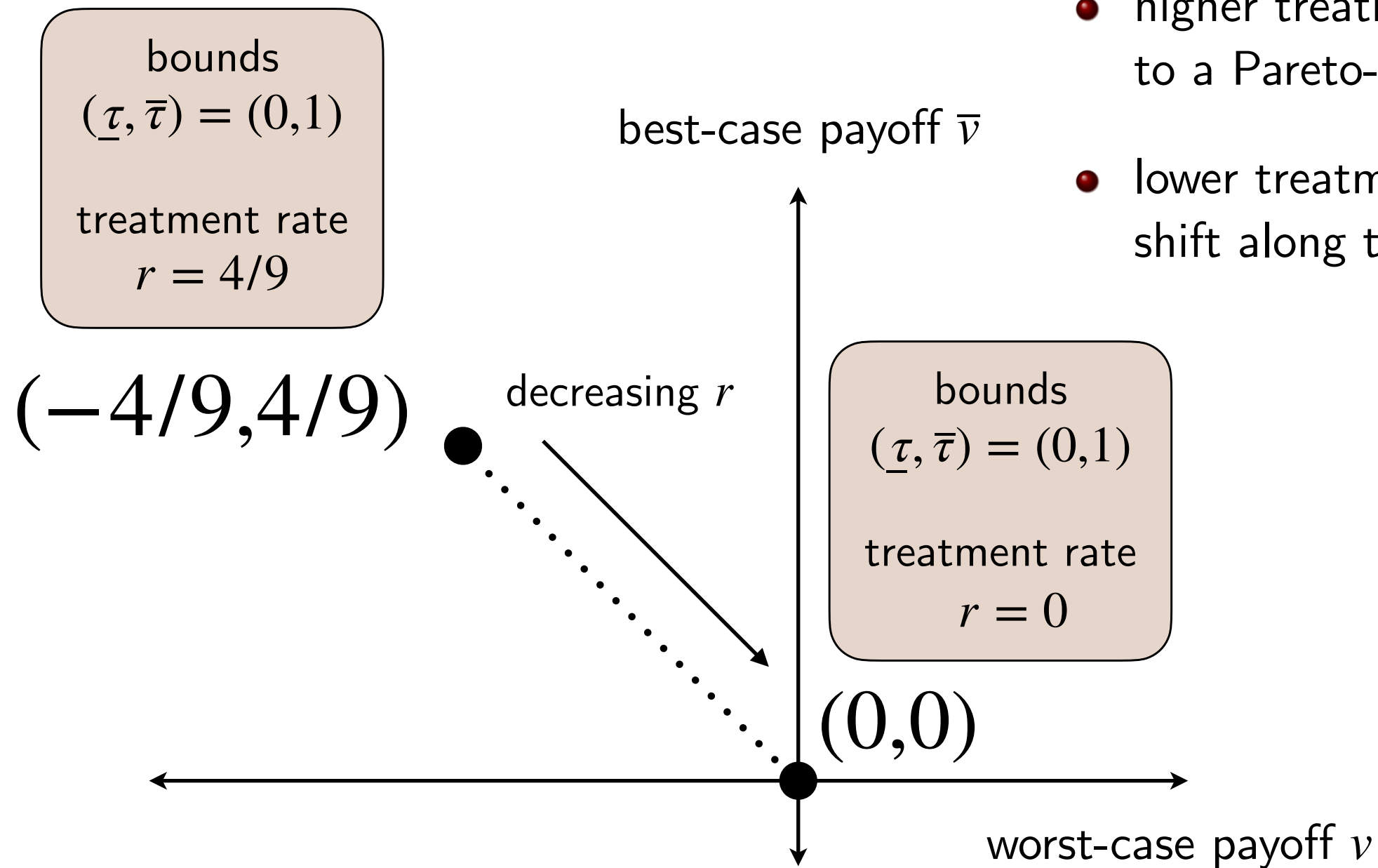
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Best-Case and Worst-Case Payoffs

Not by changing r alone

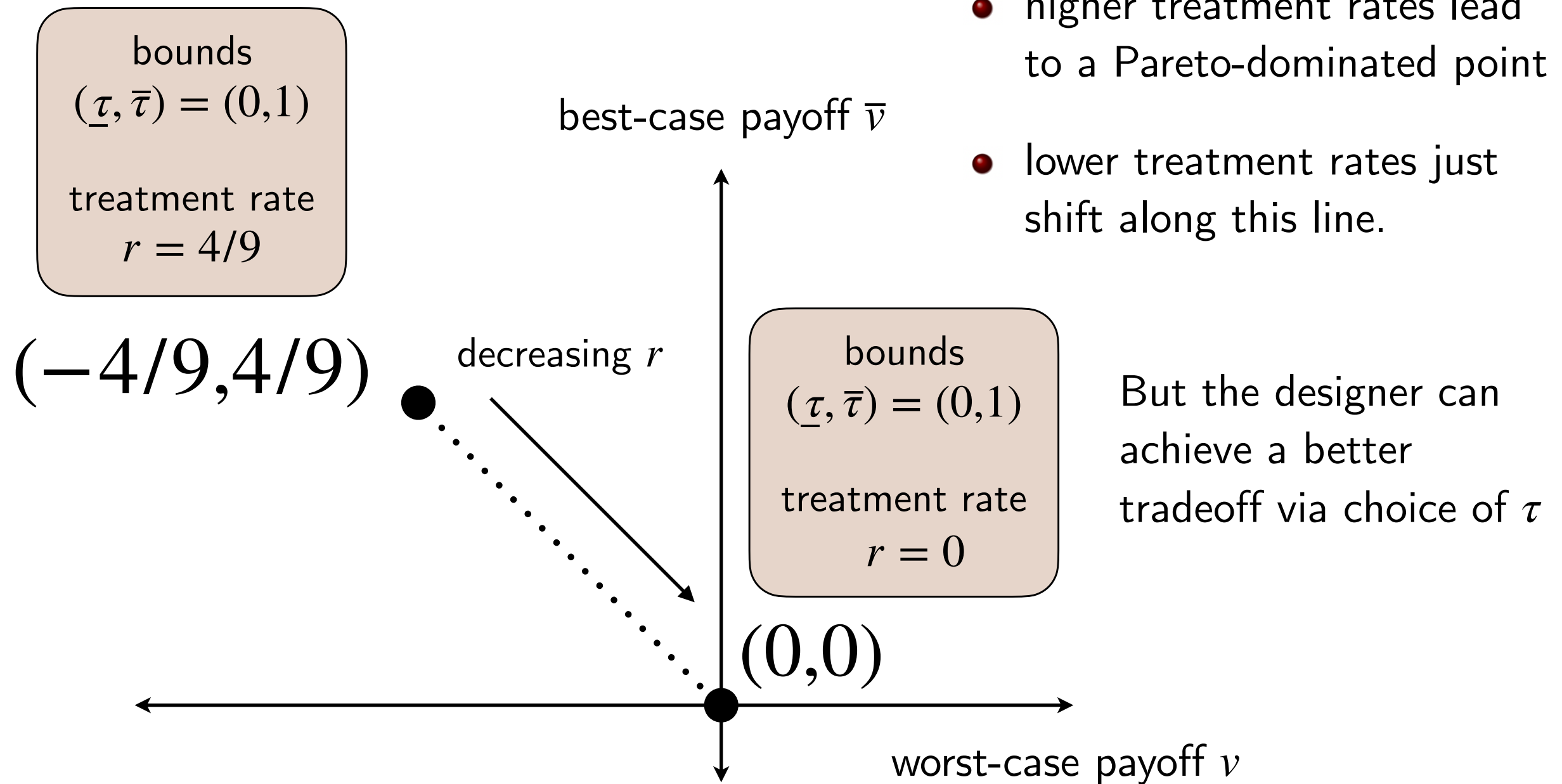
- higher treatment rates lead to a Pareto-dominated point
- lower treatment rates just shift along this line.



Best-Case and Worst-Case Payoffs

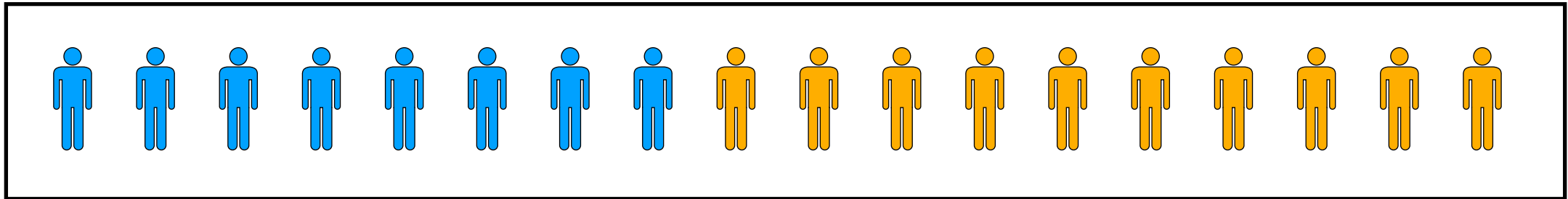
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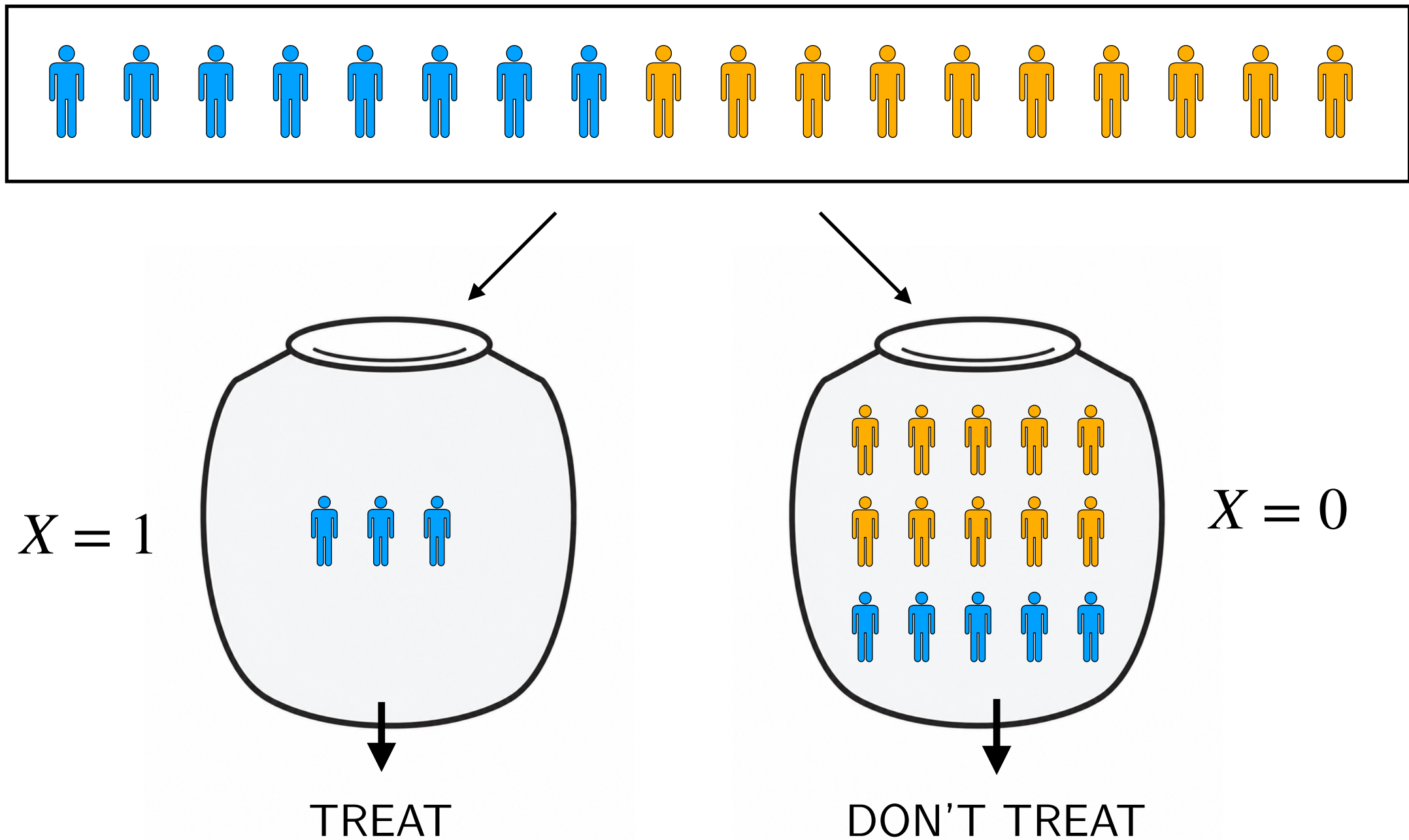
Example 1: One Group

Let $(\underline{\tau}, \bar{\tau}) = (\textcolor{red}{1}/3, 1)$ and $r = 3/18$.



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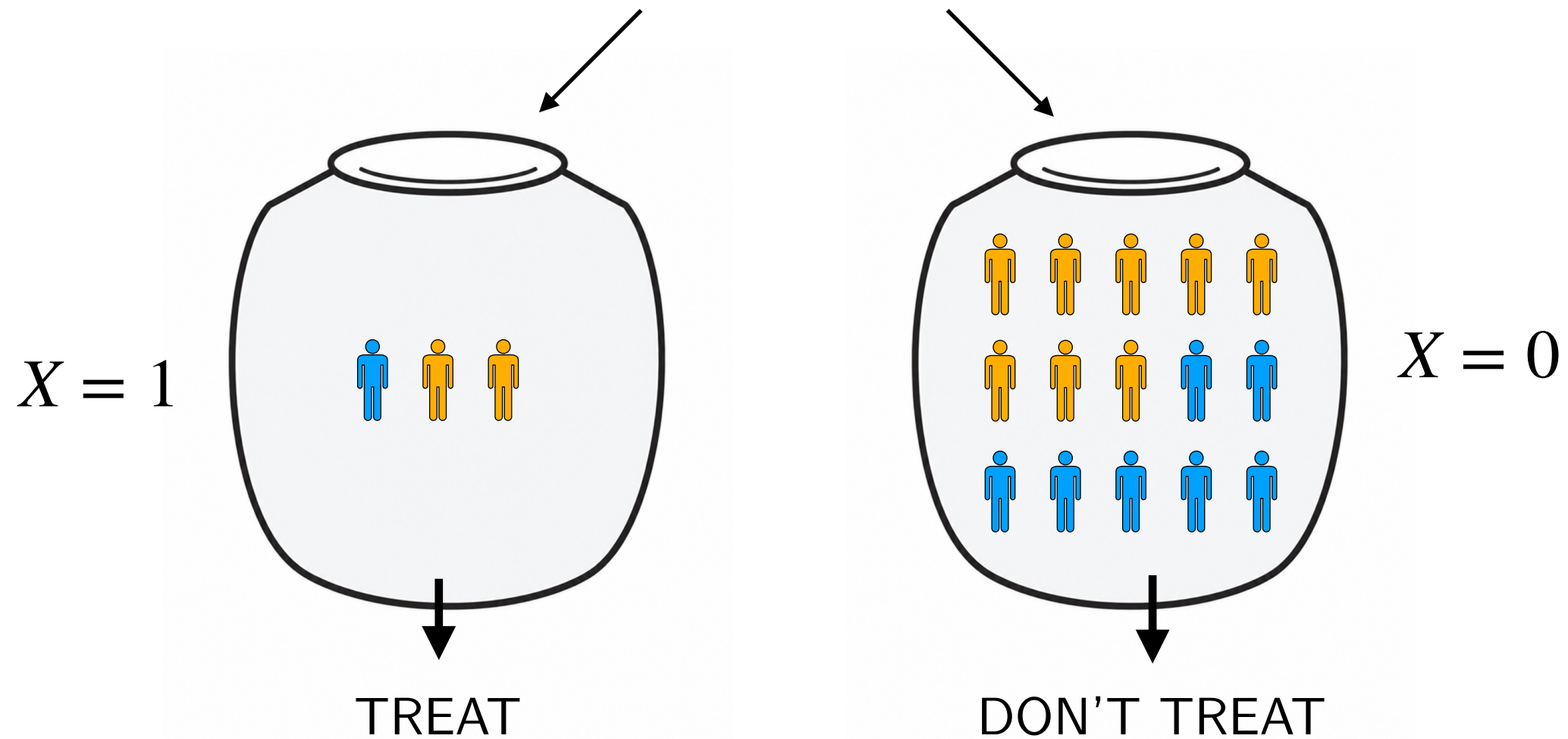
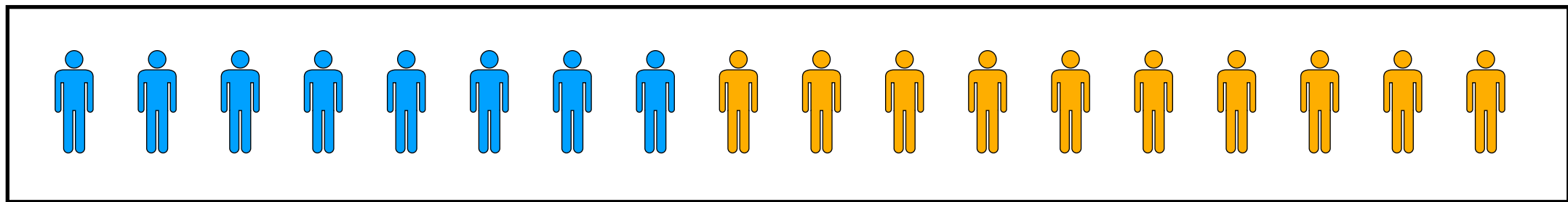
Let $(\underline{\tau}, \bar{\tau}) = (\textcolor{red}{1/3}, 1)$ and $r = 3/18$.



Treat 3/18 of patients, all of whom benefit: payoff = **3/18**

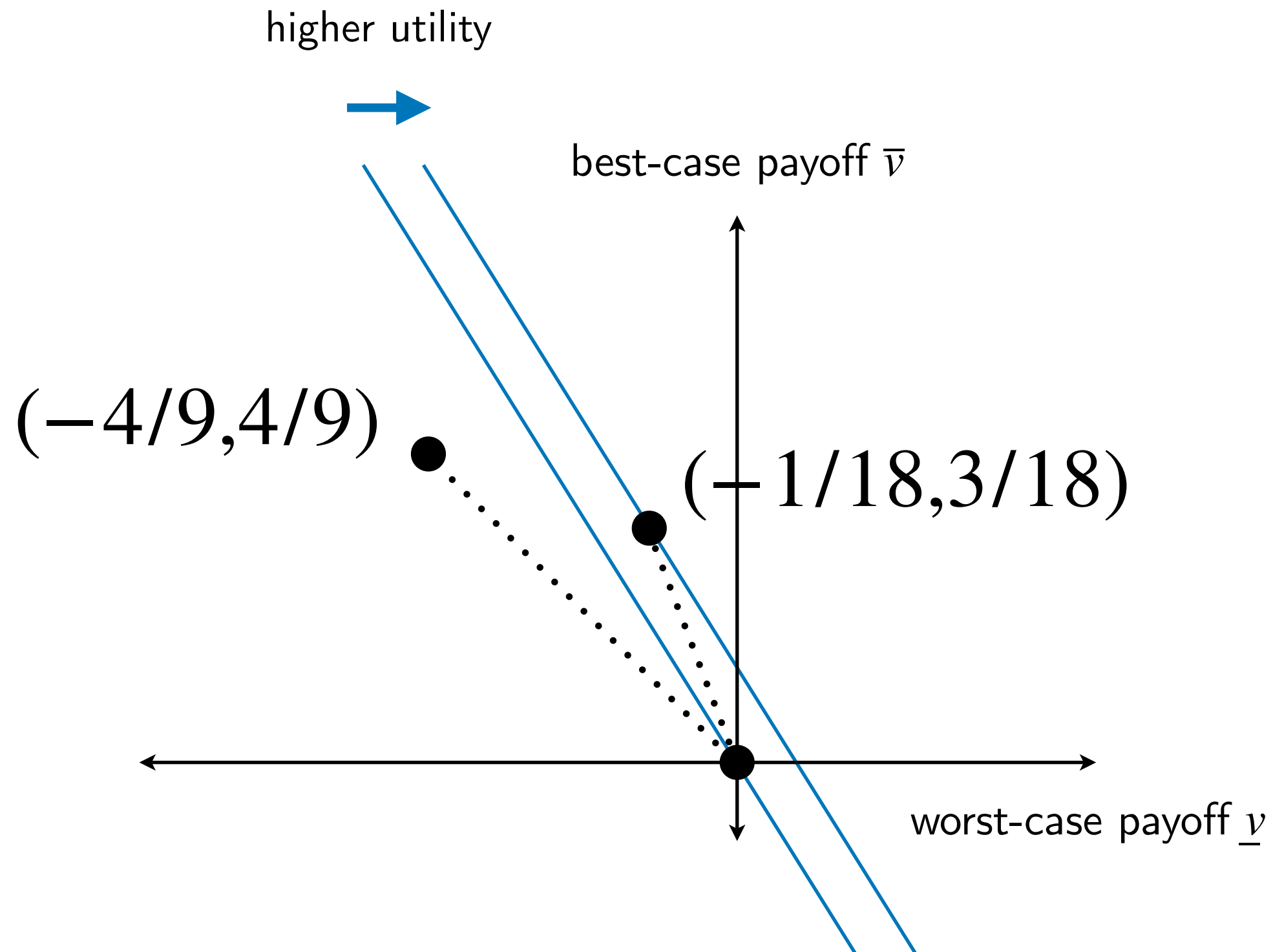
Example 1: One Group

Let $(\underline{\tau}, \bar{\tau}) = (\textcolor{red}{1/3}, 1)$ and $r = 3/18$.



Treat $3/18$ of patients, $1/3$ of whom benefit: payoff = **$-1/18$**

Best-Case and Worst-Case Payoffs

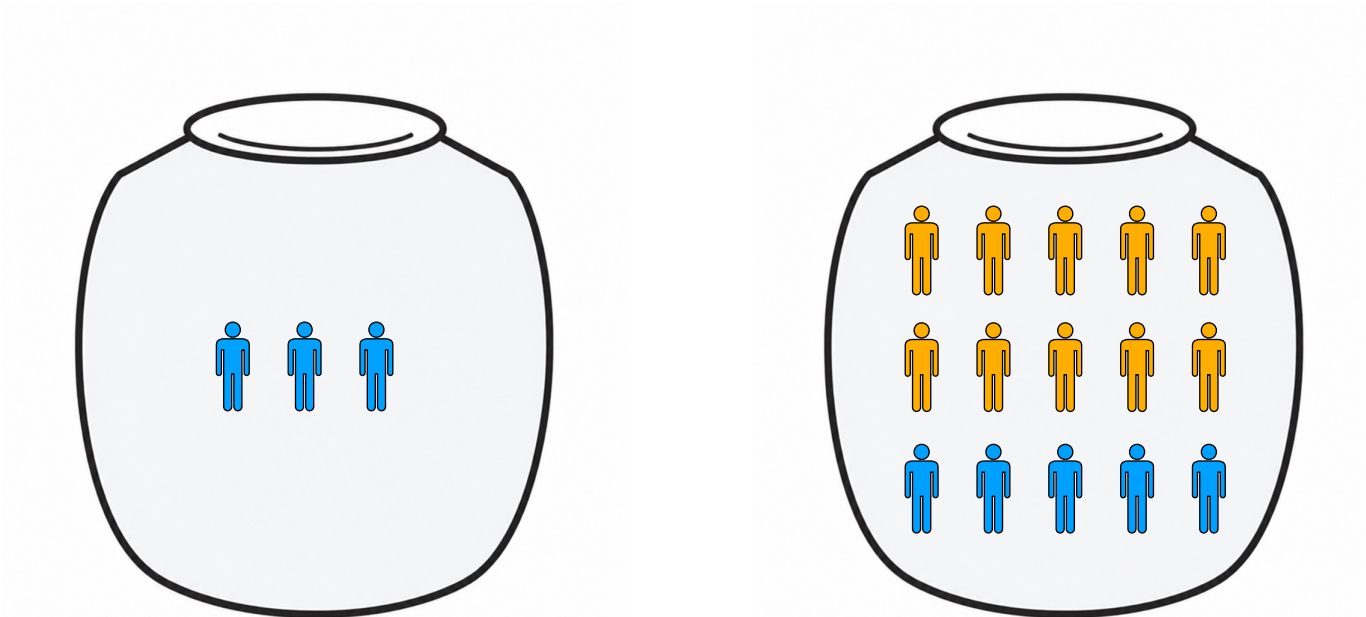


Intuition

- The aligned AI wants to identify patients who benefit from treatment. **Allow it** (set $\bar{\tau} = 1$).
- The misaligned AI wants to identify patients who are harmed by treatment. **Constrain this** (set $\underline{\tau} > 0$).

Intuition

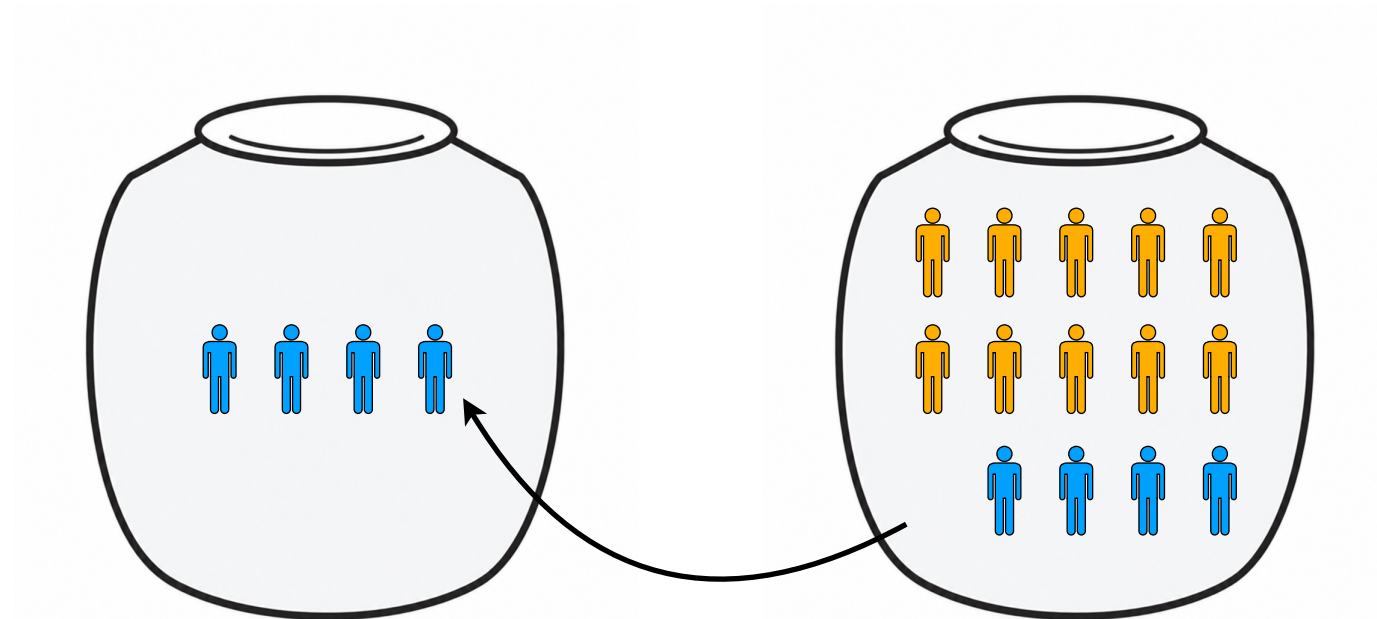
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- But the latter has a cost—limits how many patients who benefit from treatment can be correctly identified



Recall the best-case distribution with $(\underline{\tau}, \bar{\tau}) = (1/3, 1)$

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This is no longer
consistent with $\underline{\tau} = 1/3$

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- Hence pick $\underline{\tau}$ to optimally tradeoff between
 - improving the best case/worst case **tradeoff** (larger $\underline{\tau}$)
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- Hence pick $\underline{\tau}$ to optimally tradeoff between
 - improving the best case/worst case **tradeoff** (larger $\underline{\tau}$)
 - improving the **best case payoff** (smaller $\underline{\tau}$)
- The optimal treatment rate r is the largest mass of patients that can be assigned to the higher conditional probability $\bar{\tau}$

Group-Optimal Design

Result: The designer optimally sets $\bar{\tau}_g = 1$ if $p_g < 1/2$ and $\underline{\tau}_g = 0$ if $p_g > 1/2$. The other endpoint is optimally set to its extreme when $\eta = 0$, and monotonically moves towards p_g as η grows larger.

- Paper provides exact expressions

The Extensive Margin of Reliance Across Groups

Many groups

- Now allow for many groups, recalling that they are differentiated by

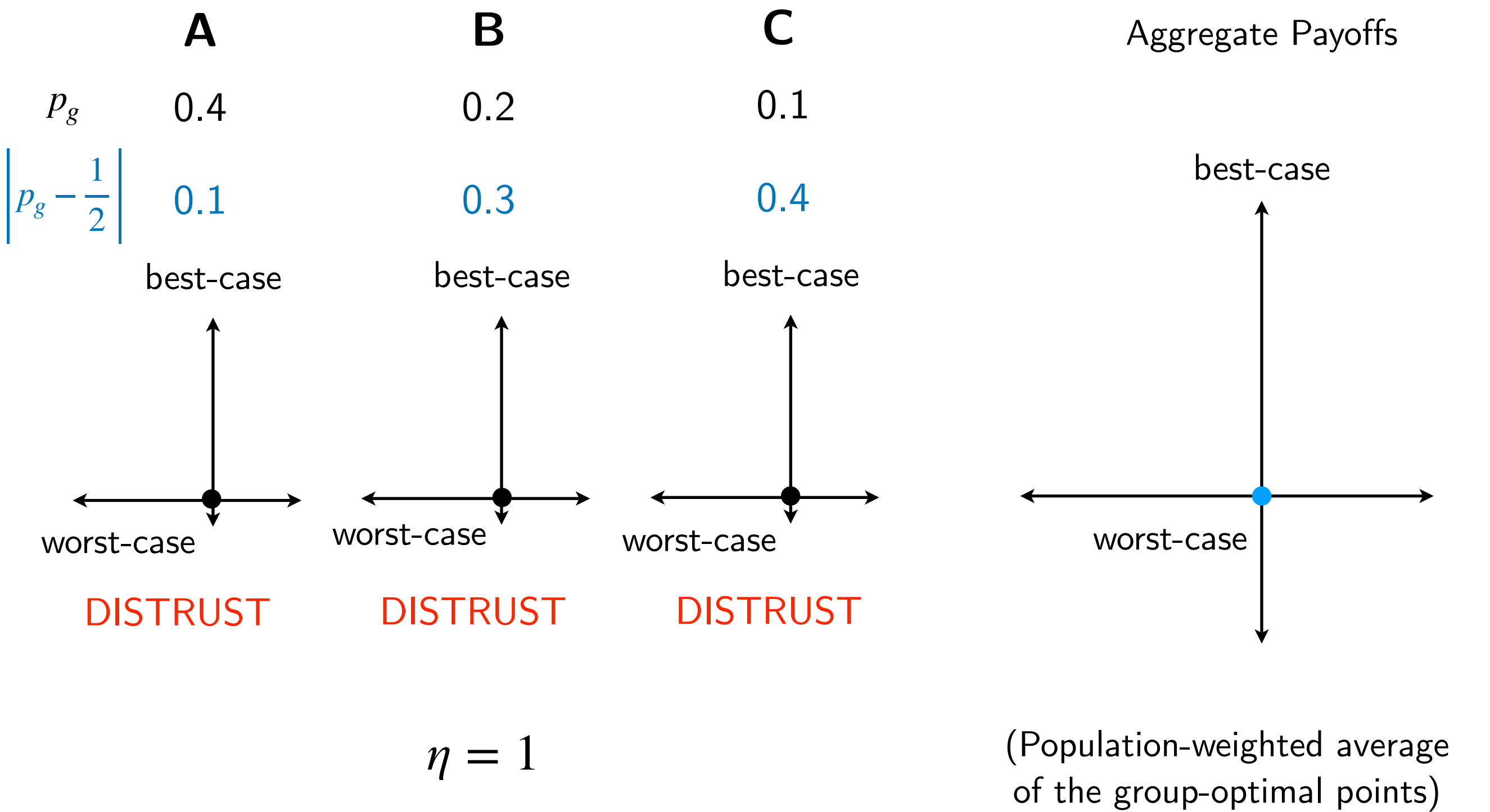
$$p_g = P(Y = 1 \mid G = g)$$

- For each group, the optimal designer choices are as described before, but how does reliance on the AI differ across these groups?
- **Result:** Groups are ordered to be increasing in $|p_g - 1/2|$.
 - Every designer relies on the AI for the first groups in this order and distrusts the AI otherwise.
 - The switching point is monotonically decreasing in η (weight on worst case payoff).

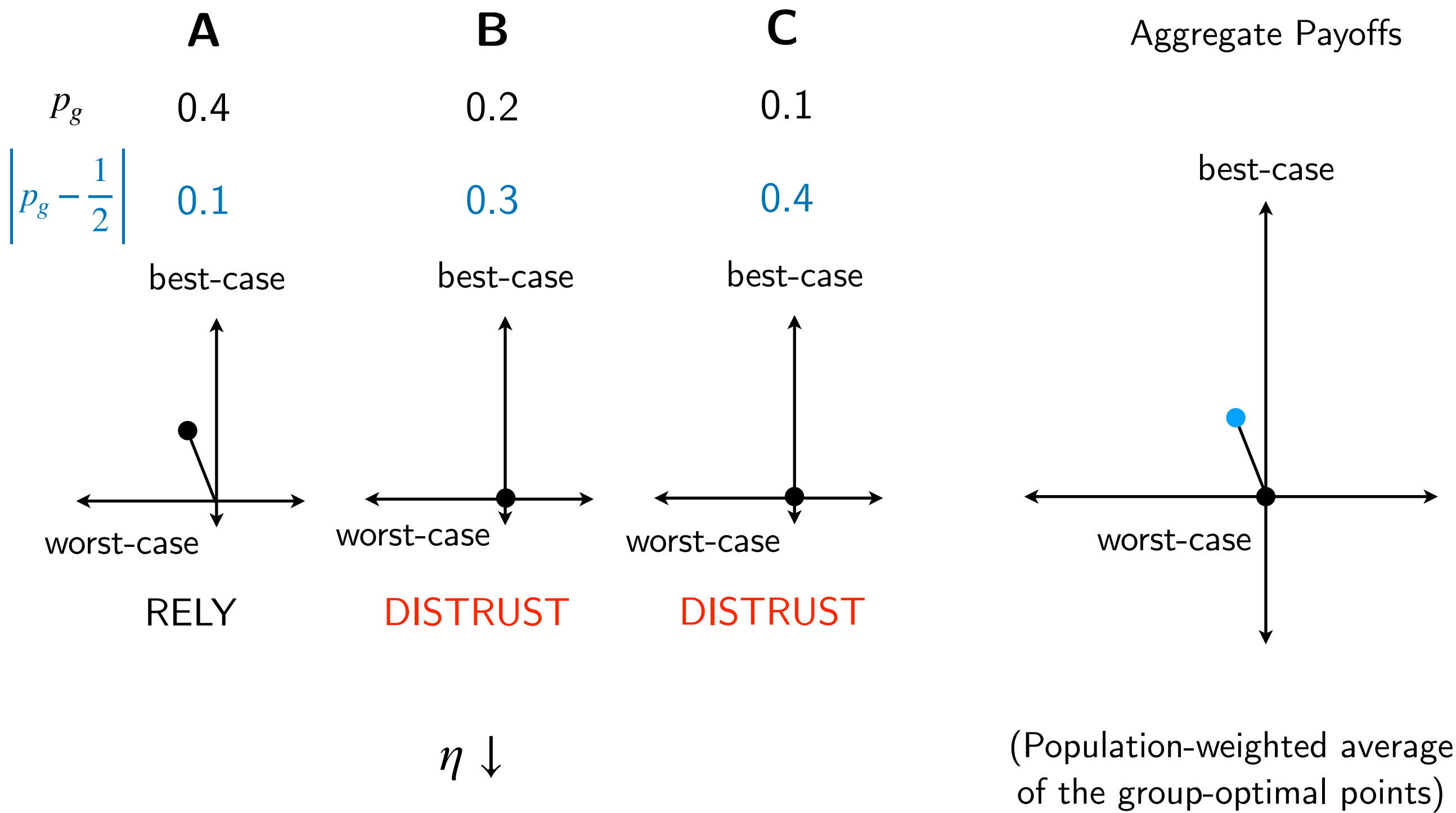
Example 2: Multiple Groups

	A	B	C
p_g	0.4	0.2	0.1
$\left p_g - \frac{1}{2}\right $	0.1	0.3	0.4

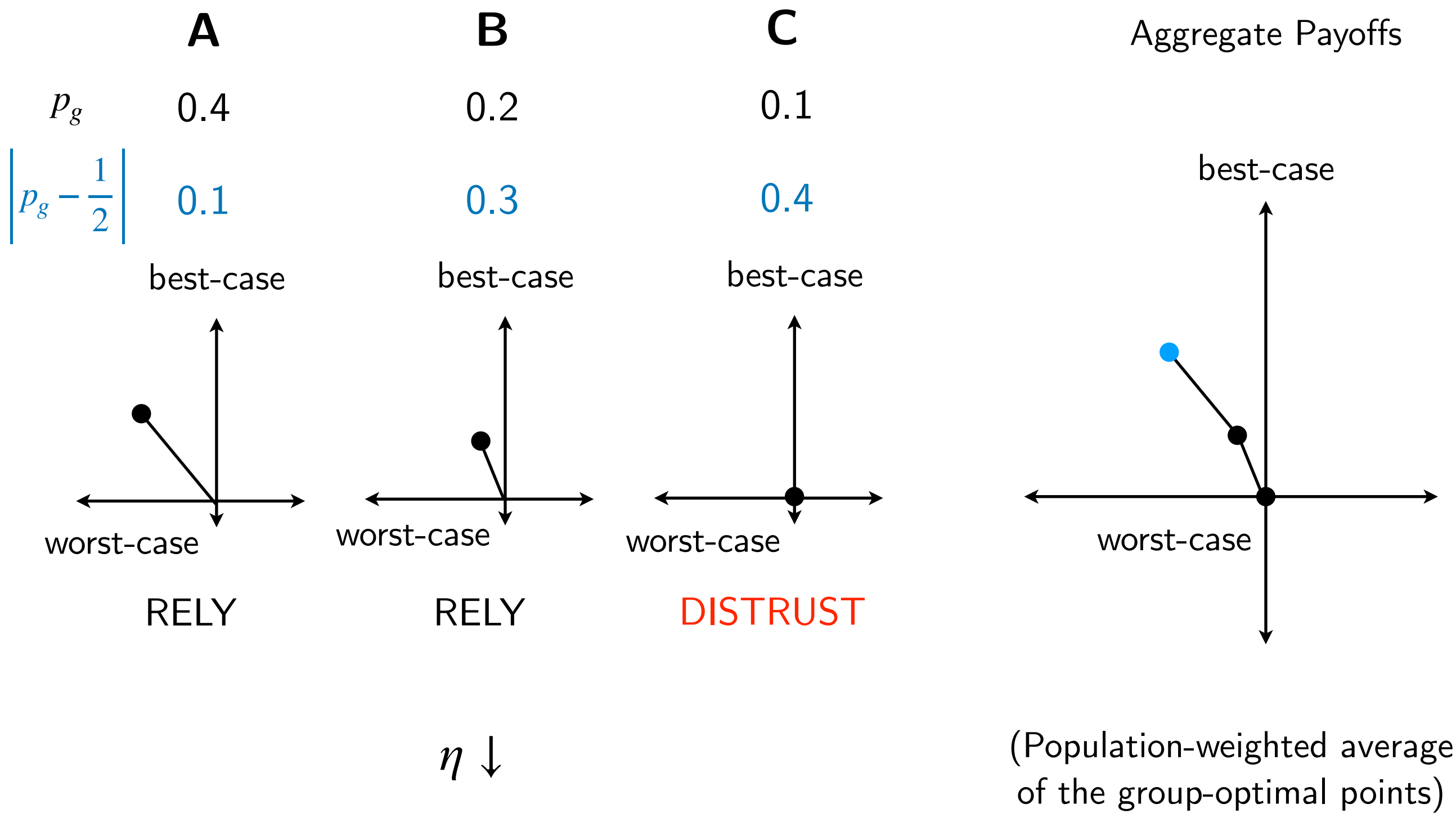
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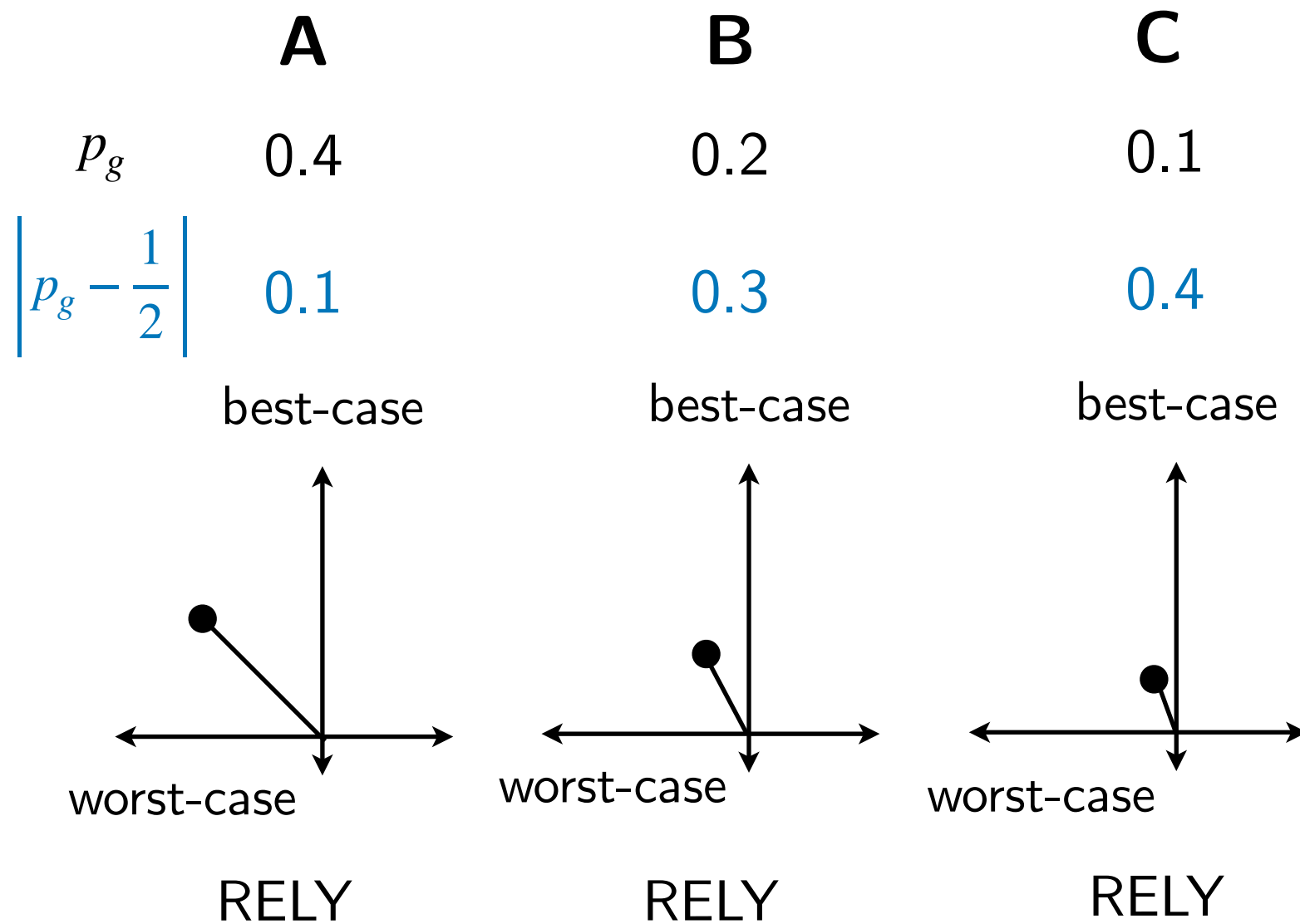
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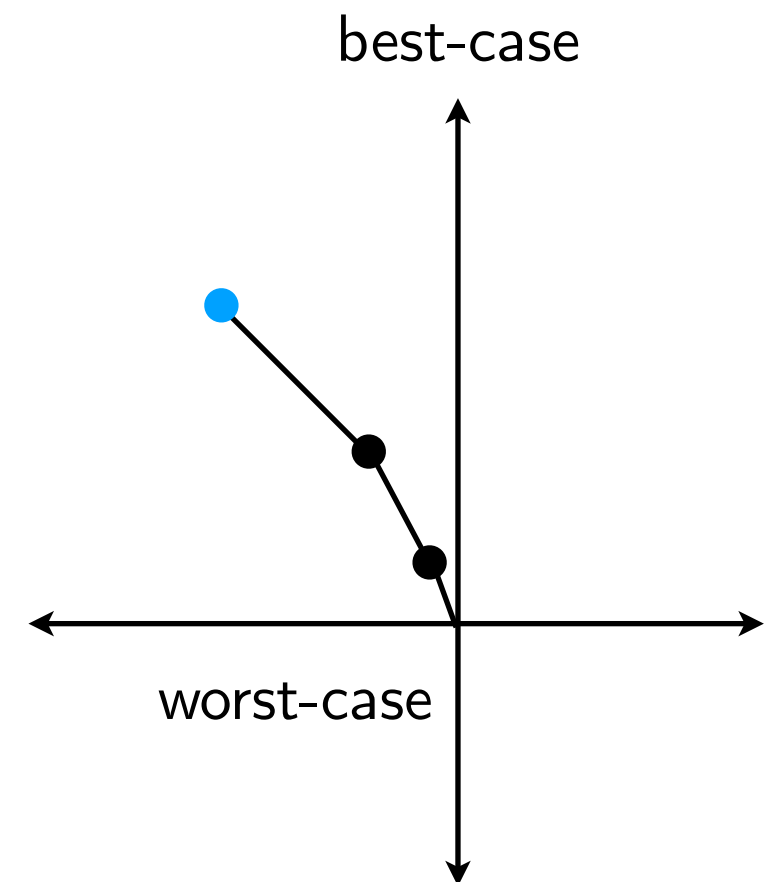


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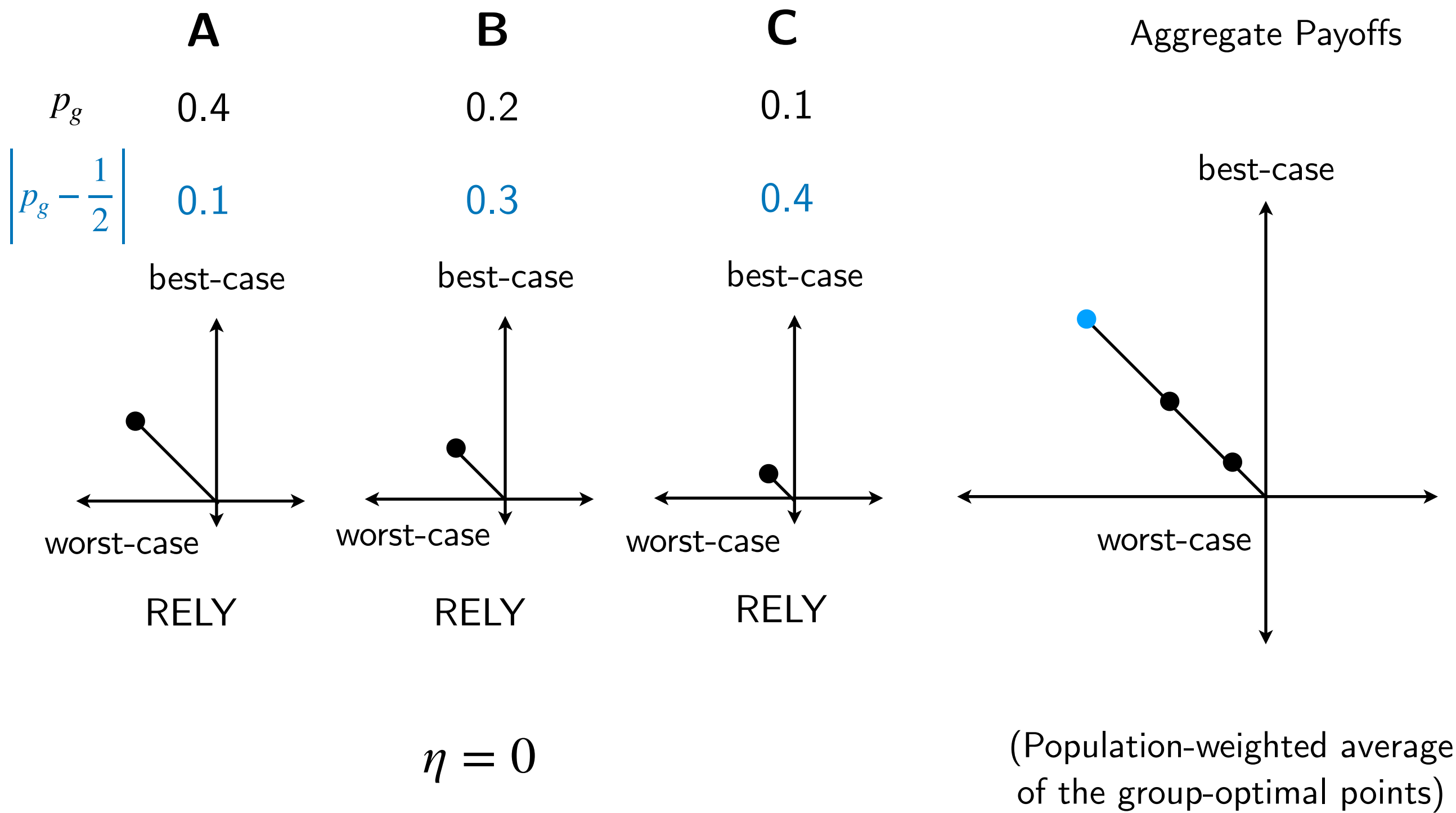
$\eta \downarrow$

Aggregate Payoffs

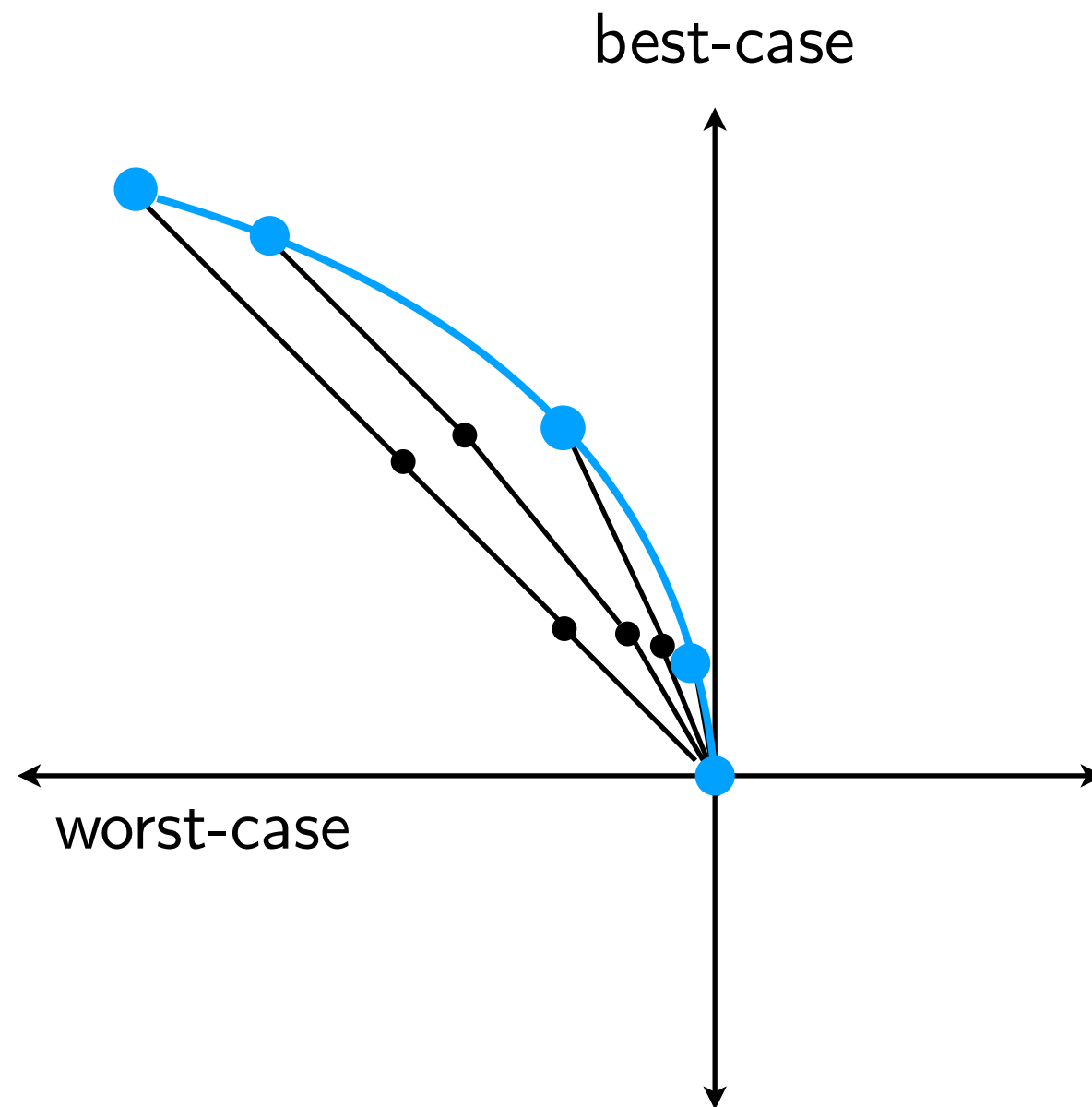


(Population-weighted average of the group-optimal points)

Example 2: Multiple Groups



Example 2: The Efficient Frontier



Takeaways

- Treatment-rate limits alone cannot improve the best-case/worst-case tradeoff
- Input restrictions can improve the tradeoff
 - Don't simply make the AI less informed, but rather control what it can be informed about
 - Allow inputs that identify treatment need but not harm (or the reverse)
- Rely on the AI first in groups where treatment success and harm are most balanced, and expand to more skewed groups as tolerance for risk grows.

Thank You